Thermodynamics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

15 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Thermodynamics: worked solutions

Solution 1 — Read the state. Return →

Convert: $P = 10^5$ Pa and $V = 0.025$ m³. $T = PV/(nR) = 2500/8.31446 = \mathbf{300.7\ K}$. One mole contains $6.02214076 \times 10^{23}$ **molecules**.

Check. Litres and kilopascals must be converted before using SI R.

Solution 2 — Isothermal expansion. Return →

For an ideal gas at fixed T, $\Delta U = 0$. $W = nRT\ln 2 = 8.31446 \times 300 \times 0.693147 = \mathbf{1729\ J}$. Therefore $Q = W = \mathbf{1729\ J}$.

Check. The gas does positive work and must absorb heat to keep its temperature fixed.

Solution 3 — Constant-pressure heating. Return →

$W = nR\Delta T = \mathbf{831.4\ J}$. $\Delta U = nC_V\Delta T = 1.5R(100) = \mathbf{1247.2\ J}$. $Q = \Delta U + W = nC_P\Delta T = \mathbf{2078.6\ J}$.

Check. More heat is needed than at fixed volume because the gas expands.

Solution 4 — Expansion without heat. Return →

$T_2 = T_1(V_1/V_2)^{\gamma-1} = 300/2^{2/3} = \mathbf{189.0\ K}$. With $Q = 0$, $W = -\Delta U = nC_V(T_1 - T_2) = 1.5R(111.0) = \mathbf{1384\ J}$.

Check. Positive work comes from internal energy, so the gas cools. The adiabatic relation used here requires reversibility.

Solution 5 — A rigid vessel. Return →

At fixed V and n, $P \propto T$, so $T_2 = \mathbf{600\ K}$. No boundary work is done. $Q = \Delta U = 1.5R(300) = \mathbf{3741\ J}$.

Check. Pressure can rise without expansion when heat is added.

Solution 6 — A rectangular engine. Return →

Idea. Net work is the rectangle's area: $(300 - 100)\,\text{kPa} \times (30 - 10)\,\text{L} = \mathbf{4000\ J}$. For monatomic gas, constant-pressure legs have $Q = \frac{5}{2}P\Delta V$ and constant-volume legs $Q = \frac{3}{2}V\Delta P$.

The four heats are $+15000$, -9000, -5000, $+3000$ J. Thus $Q_{\text{in}} = 18000$ J and $\eta = 4000/18000 = \mathbf{22.2\%}$.

Check. The heats sum to 4000 J, matching net work. Do not divide by 4000 J: that would incorrectly give 100% efficiency.

Solution 7 — A bare planet. Return →

$T = [(1 - A)F/(4\sigma)]^{1/4}$. For $A = 0.3$, **254.6 K**; for $A = 0.5$, **234.0 K**. Emitting over only half the area raises the uniform dayside temperature by $2^{1/4}$, giving **302.8 K** for $A = 0.3$.

Check. Higher albedo cools; a smaller emitting area warms. These are effective temperatures, not weather forecasts.

Solution 8 — One ideal greenhouse layer. Return →

Let the surface emit $S = \sigma T_s^4$. The layer absorbs S and emits $S/2$ upward and $S/2$ downward. Top-of-atmosphere balance gives $S/2 = \sigma T_{\text{eff}}^4$, hence $T_s = 2^{1/4} T_{\text{eff}}$. For 254.6 K, $T_s = 302.8$ **K**.

Check. The surface receives stellar energy plus downward atmospheric radiation. Real atmospheres have spectral windows, convection, clouds and vertical structure; one opaque layer is not a climate model.

Solution 9 — Mixing water. Return →

Equal masses and equal specific heat make the heat lost by the hot water equal the heat gained by the cold water. The final temperature is the average, **50°C**.

Check. The result lies between the initial temperatures. Unequal masses would require a mass-weighted average.

Solution 10 — Heat through a wall. Return →

$\dot{Q} = kA\Delta T/\ell = 0.20(0.010)(100)/0.05 = \mathbf{4.0\ W}$. This is steady conduction through the stated slab.

Check. A thicker slab reduces the power; a larger area or temperature difference increases it.

Solution 11 — An atmospheric ruler. Return →

$H = k_B T/(\mu m_u g) = 1.380649 \times 10^{-23}(250)/(28.97 \times 1.660539 \times 10^{-27} \times 9.81) = \mathbf{7.31\ km}$. At $z = H$, $P/P_0 = e^{-1} = \mathbf{0.368}$.

Check. Hotter gas has a larger scale height; heavier molecules or stronger gravity make the atmosphere more compressed.

Solution 12 — Could contraction power the Sun? Return →

$GM_\odot^2/(R_\odot L_\odot) = 9.91 \times 10^{14}$ s $= \mathbf{3.14 \times 10^7\ yr}$. A structure-dependent coefficient changes this by a factor of order unity. Nuclear fusion supplies a much larger reservoir, allowing a main-sequence lifetime of billions of years.

Check. The estimate is a reservoir divided by a rate. It is relevant to contraction phases, not the present solar energy source.

Solution 13 — A photon random walk. Return →

With $\ell = 0.01$ m, $N \sim (7 \times 10^{10})^2 = 4.9 \times 10^{21}$. Time $t = N\ell/c = L^2/(\ell c) = \mathbf{1.63 \times 10^{11}\ s} \approx 5.2 \times 10^3$ yr.

Check. This is far longer than straight-line crossing. Real opacity, density and energy exchange vary with depth, and photons are absorbed and re-emitted; the constant-mean-free-path estimate is deliberately crude.

Solution 14 — Is the atmosphere stable? Return →

The dry adiabatic lapse rate is $g/c_p = 0.00977$ K/m $= 9.77$ K/km. The parcel becomes **290.23 K**; the environment becomes **293.5 K**. At the same pressure, the colder parcel is denser and tends to sink: **stable against dry convection**.

Check. Compare parcel and environment at the same height. Moisture and condensation can change the parcel's lapse rate.

Solution 15 — A blackbody dust grain. Return →

Idea. Equate absorbed stellar power to emitted thermal power. The stellar flux at r is $\sigma T_*^4 (R_*/r)^2$. Thus $\pi s^2 \sigma T_*^4 (R_*/r)^2 = 4\pi s^2 \sigma T_d^4$, giving $T_d = T_* \sqrt{R_*/(2r)}$.

At 1 AU, $T_d = 278.3$ **K**. Since $T_d \propto r^{-1/2}$, the 150 K distance is $(278.3/150)^2 = \mathbf{3.44\ AU}$.

Check. Grain size cancels only for ideal blackbody absorption and emission. Small real grains can be hotter because their emissivity depends on wavelength.